

POINTE DU RAZ, France

WMO: **071030**Lat: **48.0389N**Lon: **4.7319W**Elev: **292**StdP: **14.54**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	33.4	35.6	23.4	17.8	35.4	26.3	20.5	37.8	52.5	47.5	49.2	47.5	15.7	70	0.804

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	72.5	63.9	69.3	62.7	67.3	61.9	65.6	69.9	64.1	67.3	63.0	65.7	11.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	89.7	67.0	62.8	86.3	65.4	61.8	83.4	64.4	30.6	70.1	29.4	67.5	28.7	65.9	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 45.8	41.0	36.3		28.7	78.3	2.2	4.3	27.1	81.4	25.7	83.9	24.5	86.4	22.8	89.5	
(5)			DB	25.9	67.6	2.2	2.7	24.3	69.6	23.0	71.2	21.7	72.7	20.1	74.7	(4)
			WB													(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	46.8	45.6	47.8	51.0	55.1	59.2	61.9	62.3	60.8	57.3	51.9	47.7	(6)
	DBStd	7.04	4.78	4.59	4.29	3.58	3.47	3.08	3.08	2.40	2.81	3.35	3.82	4.75	(7)
	HDD50	477	113	129	88	25	1	0	0	0	0	1	24	96	(8)
	HDD65	4044	564	543	532	419	310	176	108	93	131	239	392	537	(9)
	CDD50	1938	14	6	21	56	158	277	369	380	323	228	82	24	(10)
	CDD65	27	0	0	0	0	1	3	11	8	4	0	0	0	(11)
	CDH74	58	0	0	0	0	4	6	21	20	7	0	0	0	(12)
	CDH80	4	0	0	0	0	1	0	2	1	0	0	0	0	(13)
	WSAvg	17.7	21.5	19.1	18.5	16.9	16.5	14.5	14.2	14.1	14.6	18.5	21.9	22.7	(14)
	PrecAvg	32.2	3.8	3.1	2.2	2.5	2.1	1.7	1.9	2.0	2.1	3.3	3.9	3.7	(15)
(16) Precipitation	PrecMax	43.0	8.3	7.5	6.5	5.8	5.1	3.6	4.8	4.2	5.7	4.9	7.8	6.5	(16)
	PrecMin	23.2	1.0	0.6	0.3	0.3	0.3	0.4	0.3	0.2	0.2	0.7	0.6	0.6	(17)
	PrecStd	4.4	1.9	1.6	1.3	1.5	1.2	0.9	1.1	1.2	1.3	1.1	1.8	1.5	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.8	54.8	63.0	64.9	71.2	74.0	76.7	77.3	73.7	66.4	60.5	56.8		(19)
	0.4% MCWB	53.1	52.4	52.3	56.0	60.9	64.3	64.4	67.9	63.1	62.3	58.4	55.3		(20)
	2% DB	53.9	53.1	57.9	60.7	65.7	68.9	71.9	71.1	69.0	63.9	59.0	55.2		(21)
	2% MCWB	52.5	51.0	50.9	54.4	58.3	61.9	63.7	64.3	62.5	60.7	56.7	53.4		(22)
	5% DB	53.0	52.1	54.4	58.2	62.4	66.3	69.0	67.8	66.8	62.5	58.0	54.4		(23)
	5% MCWB	51.6	50.3	50.2	53.3	57.4	60.7	63.0	63.0	61.7	59.9	55.9	52.7		(24)
	10% DB	52.1	51.2	52.6	55.8	59.8	64.3	66.7	66.0	64.9	61.3	57.0	53.6		(25)
	10% MCWB	50.7	49.4	49.8	52.2	55.7	59.6	61.9	62.1	60.9	58.9	55.0	52.0		(26)
	0.4% WB	54.0	53.0	54.7	57.5	62.3	66.0	66.6	68.8	66.7	63.2	59.3	55.6		(27)
	0.4% MCDB	54.3	54.0	59.0	61.4	69.5	70.8	72.5	75.8	71.3	65.0	60.0	56.1		(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	53.1	52.0	52.7	55.7	59.2	63.0	64.7	65.8	64.4	61.9	57.6	54.3		(29)
	2% MCDB	53.5	52.7	55.1	58.8	64.3	67.2	69.6	69.7	66.9	63.2	58.5	54.8		(30)
	5% WB	52.2	51.1	51.6	53.9	57.8	61.2	63.4	64.0	63.1	60.8	56.5	53.4		(31)
	5% MCDB	52.6	51.9	53.2	56.8	61.5	64.8	67.5	66.4	65.0	62.0	57.6	54.0		(32)
	10% WB	51.1	49.9	50.6	52.7	56.3	60.0	62.3	62.9	62.0	59.9	55.4	52.4		(33)
	10% MCDB	51.9	51.0	52.2	55.2	59.0	63.1	65.5	65.1	63.9	61.1	56.7	53.3		(34)
(35) Mean Daily Temperature Range	MDBR	5.0	5.7	7.0	7.5	7.3	7.6	7.6	7.4	7.5	5.8	5.2	5.6		(35)
	5% DB	4.0	5.2	9.4	10.2	11.1	11.4	11.3	10.0	10.1	6.2	5.4	4.7		(36)
	5% WB	4.9	4.8	6.6	7.0	6.9	7.2	6.5	5.9	6.4	5.2	5.8	5.9		(37)
	MCDBR	3.8	4.0	6.5	8.4	9.7	9.7	8.9	8.0	5.5	4.7	4.5	4.5		(38)
	MCWBR	5.0	4.5	5.3	6.2	6.5	6.8	6.1	5.8	5.9	5.2	5.8	6.1		(39)
(40) Clear-Sky Solar Irradiance	taub	0.337	0.350	0.374	0.381	0.384	0.379	0.380	0.369	0.359	0.358	0.339	0.330		(40)
	taud	2.405	2.395	2.351	2.353	2.391	2.430	2.432	2.480	2.489	2.471	2.431	2.375		(41)
	Ebn at Noon	230	255	266	276	279	280	278	277	269	249	227	215		(42)
	Edh at Noon	22	27	33	36	36	35	35	32	29	25	21	20		(43)
(44) All-Sky Solar Radiation	RadAvg	317	582	969	1512	1783	1929	1842	1583	1265	743	398	266		(44)
	RadStd	23	56	102	121	152	201	91	125	90	55	29	29		(45)

Historical Trends